

Ankur Desai

(248) 657 – 3805 | ardusa05@gmail.com | ardusa.github.io | linkedin.com/in/ardusa

EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design (Java), Computer Organization and Architecture, Data Structures & Algorithms
- **Programming:** Java-based projects in OOP and data structures

EXPERIENCE

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Collaborated with engineering peers on software development and teamwork: eliminated manual onboarding by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, enabling real-time attendance tracking and cutting processing time by 80%.
- Enforced production-grade quality via GitHub Actions CI/CD with code review workflow and automated testing (90% coverage gates across 20+ tests), with attention to detail on coverage and review standards.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Introduced Neo4j after design discussion to offload recursive relational queries from PostgreSQL, cutting p95 latency 48% (850ms → 440ms) and eliminating SQL join overhead for multi-hop traversals.
- Refactored Cypher with relationship-type indexing (problem-solving for query performance); indexed traversals ran 3× faster than equivalent SQL self-joins and removed full-table scans on relationship lookups.
- Integrated dual-provider OAuth 2.0 (Google + GitHub) with httpOnly JWT cookies, refresh token rotation, and user-scoped graph access control.
- Implemented GitLab CI/CD pipeline (development tools for build, test, deploy); participated in code reviews via branch-protected merges and documented pipeline configuration and development best practices, enforcing 85% coverage across 35+ tests.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery* May 2025 – Nov. 2025

- Engineered multi-client agentic voice assistant that batches natural dialogue, resolves implicit speaker intent, and dispatches real tasks across concurrent voice streams.
- Reduced RESTful API response time by 73% by offloading RAG pipeline to async Redis/Celery task queue.
- Built bidirectional Electron + Next.js client with an embedded HTTP server for OAuth callback interception; real-time events via IPC and a dedicated webhook server deliver sub-50ms propagation.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow* Feb. 2025

- Won Interactive Media Track: Gesture-controlled wizard duel where full-body movements cast spells, block attacks, and trigger abilities against an AI opponent.
- Built full-body gesture pipeline using MediaPipe Pose to map 33 skeletal landmarks into game actions; optimized inference loop to sustain 60 FPS with sub-20ms input lag.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash

Frameworks Spring Boot, FastAPI, Node.js, Angular, React, Next.js, Redis

Databases PostgreSQL, Neo4j, MongoDB

Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitLab CI/CD, CI/CD Pipelines